



SERA-HAR: A real-time human activity recognition framework via staged early-exit and inter-channel attention

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Abstract

Real-time human activity recognition (HAR) must balance accuracy and latency under device constraints. We present SERA-HAR, a residual temporal convolutional framework augmented by two lightweight attentions. A staged early-exit attention reads the dispersion of temporal attention to decide when to stop observing and classify, while an inter-channel correlation attention (ICCA) captures cross-sensor dependencies to complement temporal encoding. On UCI-HAR, WISDM, and KU-HAR, SERA-HAR attains 94.6–96.7% accuracy and up to 95.95% macro-F1 within a unified time-domain pipeline. On a workstation, mean inference latency is 1.79 ms with throughput of 760 samples/s; the early-exit policy reduces expected observation time by as much as 43% relative to fixed-window processing. These results indicate a favorable accuracy–latency trade-off for wearables, healthcare monitoring, and pervasive sensing.

Keywords Real-time systems · Human activity recognition · Early-exit mechanism · Wearable sensors

1 Introduction

Sensor-based HAR uses wearable sensor data to automatically identify user behaviors, with broad applications in health monitoring, smart security, and behavior analysis. Compared to vision-based methods, wearable inertial sensors provide better privacy and robustness to environmental changes [1, 2]. These sensors generate stable multimodal signals for extracting rich motion and physiological features [3]. The ubiquity of smartphones and wearables has also led to numerous public HAR datasets and data collection frameworks, accelerating progress in the field [4].

Although current HAR systems can achieve high accuracy in offline settings, most still rely on fixed-size sliding windows for segmenting sensor streams. This rigid windowing can misalign with variable activity durations, blurring distinctive motion patterns and reducing robustness [5]. Sliding

windows also generate redundant data, increasing computation and latency, which hinders real-time responsiveness [6].

However, several challenges remain: most HAR models still use predetermined observation lengths without evaluating sufficiency [5], often ignore inter-sensor relationships in multimodal data [6], and increasing model complexity can degrade efficiency on wearable devices [7]. While large models and multimodal representations have recently been explored for sensing and behavior modeling [8], achieving low-latency inference on resource-constrained devices remains difficult [9, 10].

The main contributions of this work are:

- **SERA Module:** A staged early-exit mechanism that adaptively shortens observation based on attention uncertainty, reducing latency.
- **ICCA Module:** A channel-wise attention mechanism for modeling inter-sensor dependencies and enhancing multimodal features.
- **Real-time HAR System:** An efficient residual TCN-based framework that integrates both modules, achieving high accuracy and low latency across benchmarks.

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2 Related Work

Deep learning techniques have greatly advanced HAR by improving temporal modeling. Convolutional and recurrent neural networks can learn complex features from raw sensor streams and often outperform hand-crafted feature pipelines [11]. Residual temporal convolutional networks and hybrid CNN–LSTM or CNN–RNN designs support efficient parallel sequence processing and have shown strong performance on wearable HAR benchmarks [12, 13].

Attention mechanisms—temporal attention, channel attention and multi-head variants—help focus on informative time segments and sensor channels, improving the recognition of complex or overlapping activities [14, 15]. Early-exit networks and anytime prediction schemes aim to reduce latency by enabling the model to stop inference when sufficient confidence is reached; their utility and trade-offs for HAR have been recently investigated [16].

Researchers have proposed many practical architectures and efficiency-oriented designs. Representative works include WISNet with channel-aware convolutional blocks [17], MultiCNN-FilterLSTM that combines multi-scale filters with lightweight recurrent modules for IoT settings [13], and Multi-STMT which fuses convolutional and recurrent (or GRU-based) modules with attention for spatiotemporal modeling [18]. Optimization-driven or metaheuristic strategies have also been explored to tune architectures and training [19].

Transfer learning, few-shot learning and self-/semi-supervised methods address data scarcity and domain shifts. Surveys and recent methods summarize cross-domain personalization and practical transfer strategies [20]. Few-shot/meta-learning approaches demonstrate promising results under limited labeled data [21]. Contrastive and temporal consistency-based self-supervised learning techniques have been effective in improving representation quality; recent works combine contrastive objectives with distillation or capsule/transformer modules to further boost low-supervision performance [22, 23].

Beyond inertial sensing, other modalities and tools expand HAR capabilities. Some works transform inertial signals into spectrogram-like representations and leverage generative models to enrich features [24]; RF and through-wall sensing combined with point-based networks have been applied to non-line-of-sight action recognition [25]. Simulation frameworks (e.g., WIMUSim) help generate realistic IMU variability for robust training [26]. Finally, hybrid pipelines that combine handcrafted features and learned representations remain practical for continuous, low-power deployments [27, 28].

3 Method

3.1 Problem Overview

HAR aims to identify behaviors from continuous sensor streams collected by smartphones or wearables [28]. Despite progress brought by deep learning, most models still rely on long, fixed-size observation windows, which limits real-time adaptability in noisy and dynamic environments [29]. To address delayed response and the difficulty of recognizing short or abrupt actions, we propose **SERA-HAR**. The overall pipeline—preprocessing, temporal feature extraction, attentions, and classification—is shown in Fig 1. The following subsections describe each component.

3.2 Data Preprocessing

We adopt a unified time-domain pipeline for UCI-HAR, KU-HAR, and WISDM. For KU-HAR and WISDM, a third-order Butterworth low-pass filter (20 Hz cutoff) suppresses high-frequency noise. WISDM is resampled to 20 Hz via linear interpolation to alleviate irregular sampling. All channels are Z-score normalized using training-set statistics. We operate purely in the time domain: no spectral features are computed, and the 561 engineered features of UCI-HAR are not used.

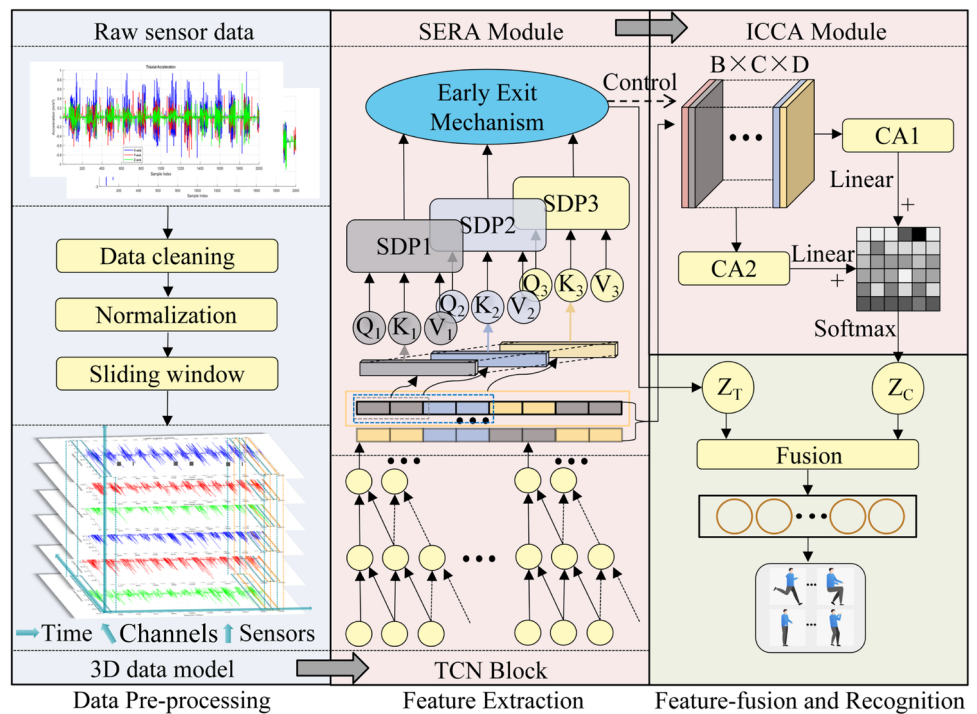
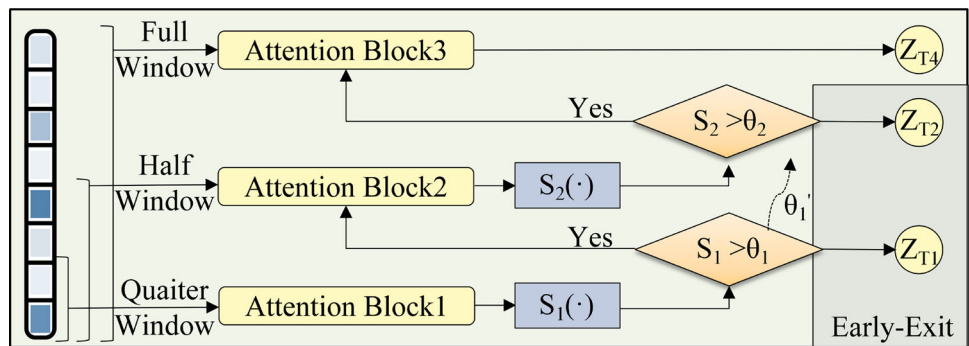
After denoising and normalization, raw streams are segmented into windows (UCI-HAR 2.56 s, WISDM 3 s, KU-HAR 3 s). Each window is split into four sub-windows (1/4, 1/2, 3/4, full). The 3/4 part is used only for feature accumulation. Decisions are made at 1/4, 1/2, and full. For a stage- i prefix the input is a time-domain tensor $\mathbf{X} \in \mathbb{R}^{C \times T_i}$, which enables progressive inference and early decisions [31, 32].

3.3 Temporal Feature Extraction using TCN

A TCN extracts robust temporal features using dilated *causal* convolutions to capture short/long-term dependencies without future leakage [17]. Each residual block consists of Conv1d–Norm–ReLU with dilation, plus dropout and a 1×1 residual projection when needed. The TCN produces stage-wise prefix features that feed the attention modules.

3.4 Staged Early-exit Recognition Attention

Each window is partitioned into four equal sub-windows that share the same label. Early decisions are allowed only at the quarter and half prefixes and at the full window; the 3/4 slice is used *only* to accumulate features. Let the stage index $i \in \{1, 2, 3\}$ correspond to the 1/4, 1/2, and full prefixes. At stage i , the observed prefix $X_{1:i}$ is first encoded by the TCN to produce a *prefix feature* $\tilde{Z}_i$. The attention block then takes $\tilde{Z}_i$ and returns a stage representation z_i together with a temporal

Fig. 1 Overall architecture of SERA-HAR**Fig. 2** Illustration of the Early-Exit Mechanism

attention distribution $\alpha^{(i)}$ over the currently observed time steps.

We summarize the dispersion of $\alpha^{(i)}$ by variance and entropy:

$$\sigma_i^2 = \text{Var}(\alpha^{(i)}), \quad H(\alpha^{(i)}) = -\sum_k \alpha_k^{(i)} \log \alpha_k^{(i)}. \quad (1)$$

The stage score is a learnable fusion of the two:

$$S_i = \lambda_1 \sigma_i^2 + \lambda_2 H(\alpha^{(i)}), \quad (2)$$

with λ_1, λ_2 learned jointly with the network.

Given a threshold θ , the decision at stages $i \in \{1, 2\}$ is

$$\begin{aligned} S_i < \theta &\Rightarrow \text{exit at stage } i, \\ S_i \geq \theta &\Rightarrow \text{append next sub-window and continue.} \end{aligned} \quad (3)$$

Stage 3 always outputs the final prediction.

To keep the decision threshold positive and numerically stable, we reparameterize

$$\theta = \text{softplus}(\tilde{\theta}) = \log(1 + e^{\tilde{\theta}}). \quad (4)$$

We optimize $\tilde{\theta}$ jointly with network weights and add a small quadratic regularizer to discourage indecisive scores around the boundary:

$$\mathcal{L}_{\text{exit}} = \eta \frac{1}{M} \sum_{i=1}^M (S_i - \theta)^2. \quad (5)$$

The training objective becomes

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{exit}}. \quad (6)$$

Initialization uses $\theta_0 = \text{median}\{S_i\}$ measured during the first few epochs. After training, a short validation sweep selects θ^*

Fig. 3 ICCA during staged inference. The box *Feature* is the TCN-encoded *prefix feature* $\tilde{Z}_i$ from the current prefix $X_{1:i}$. At each stage $i \in \{1/4, 1/2, \text{full}\}$, ICCA maps $\tilde{Z}_i$ to Q, K, V and computes channel attention to produce Z_{ICCA} (Eq. (7)). Solid arrows indicate *continue* ($S_i \geq \theta$; append next sub-window); dashed arrows indicate *exit* ($S_i < \theta$; classify with current (z_i, Z_{ICCA}))

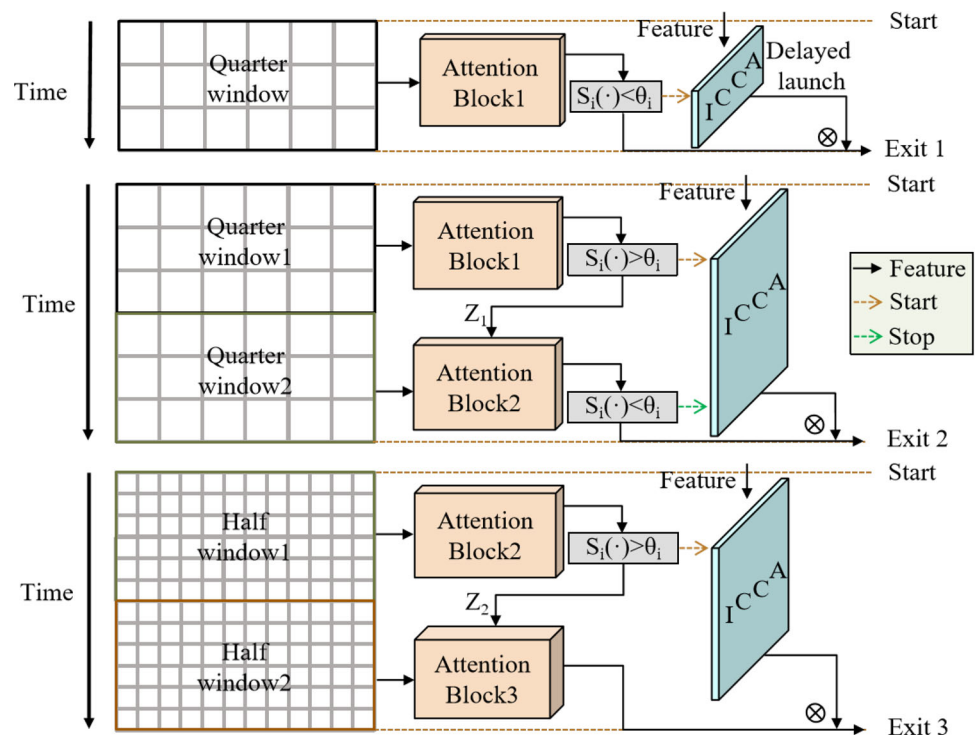


Table 1 Overview of datasets and preprocessing

Dataset	Sampling rate	Window length	Sliding step	Sub-windows
UCI-HAR	50 Hz	2.56 s	50% overlap	1/4, 1/2, 3/4, full
WISDM	20 Hz	3.0 s	50% overlap	1/4, 1/2, 3/4, full
KU-HAR	100 Hz	3.0 s	0% overlap	1/4, 1/2, 3/4, full

that maximizes early exits while keeping the accuracy drop within 0.2%. The full module is illustrated in Fig 2. Tab. 1

3.5 ICCA-based Channel-Wise Attention

Temporal evidence alone can under-utilize cross-sensor structure. The ICCA augments the stage prefix with channel-wise self-attention. Let $X \in \mathbb{R}^{B \times C \times D}$ denote the stage- i prefix embedding (batch B , channels C , temporal embedding D). ICCA computes

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right), \quad Z_{\text{ICCA}} = AV, \quad (7)$$

with $Q, K, V \in \mathbb{R}^{C \times d}$ learned projections. A captures sensor-wise dependencies and Z_{ICCA} enhances spatial features. ICCA is synchronized with SERA: it is recomputed on the current prefix and halts once an early exit is triggered, ensuring temporal-spatial alignment. The resulting maps evolve with observation depth, as visualized in Fig 3.

3.6 Feature Fusion and Classification

Let $Z_T, Z_C \in \mathbb{R}^{B \times D}$ be the pooled outputs from the SERA and ICCA paths. A shared fully connected layer produces scores

$$\begin{aligned} s_T &= \text{FC}(\text{AvgPool}(Z_T)), \\ s_C &= \text{FC}(\text{AvgPool}(Z_C)). \end{aligned} \quad (8)$$

Softmax weights

$$\gamma_T = \frac{e^{s_T}}{e^{s_T} + e^{s_C}}, \quad \gamma_C = \frac{e^{s_C}}{e^{s_T} + e^{s_C}}, \quad (9)$$

yield the fused representation

$$Z_{\text{fused}} = \gamma_T Z_T + \gamma_C Z_C, \quad (10)$$

which is fed to the classifier.

Table 4 Comparison on three datasets

Model	UCI-HAR Acc. (%)	F1 (%)	WISDM Acc. (%)	F1 (%)	KU-HAR Acc. (%)	F1 (%)
MLP	85.34 ± 0.42	90.42 ± 0.38	90.48 ± 0.55	90.90 ± 0.49	–	–
CNN (1D)	92.71 ± 0.33	92.23 ± 0.35	–	–	–	–
Res-LSTM	94.23 ± 0.28	94.51 ± 0.30	–	–	–	–
Transformer	94.00 ± 0.31	94.00 ± 0.29	–	–	–	–
LSCG	94.33 ± 0.27	94.51 ± 0.33	–	–	–	–
CNN-LSTM	–	–	93.65 ± 0.44	92.88 ± 0.41	94.95 ± 0.52	92.89 ± 0.47
TCN-Attention	–	–	93.01 ± 0.48	92.96 ± 0.42	95.91 ± 0.59	93.45 ± 0.55
Bi-GRU	–	–	95.48 ± 0.53	95.40 ± 0.50	–	–
Bi-GRU+CNN	–	–	–	–	95.92 ± 0.61	95.48 ± 0.57
CNN+ResNet+BiRNN	–	–	97.48	95.95	–	–
SERA-HAR	94.63 ± 0.18	94.66 ± 0.20	96.65 ± 0.52	95.71 ± 0.47	95.47 ± 0.71	95.95 ± 0.65
<i>Recent literature (reported)</i>						
Hybrid CNN+Channel Attn. [28]	98.92	–	98.80	–	–	–
WISNet [17]	96.41	95.00	–	–	–	–
Multi-STMT [18]	–	–	–	–	97.99	–

Unmarked rows are from our unified reproduction (mean ± std over 5 runs), while cited rows are directly taken from the corresponding publications.

4.4 Ablation Study

Table 5 reports the ablation analysis on UCI-HAR and WISDM, with results averaged over five runs. Removing either SERA or ICCA substantially reduces performance, highlighting their complementary roles. Moreover, single-stage variants show that very short contexts (1/4 or 1/2) are insufficient, while using only the full window restores accuracy but sacrifices timeliness.

4.5 Real-Time Study

4.5.1 Theoretical Time Savings

Let c_i denote MACs for stage i (quarter/half/full) and p_i the probability of exiting at that stage. The expected compute is

$$\mathbb{E}[\text{MACs}] = \sum_{i=1}^3 p_i \sum_{j=1}^i c_j, \quad \sum_{i=1}^3 p_i = 1. \quad (11)$$

As shown in Table 6, staged inference yields substantial expected observation-time savings relative to a fixed-window pipeline.

Table 5 Ablation on UCI-HAR and WISDM

Variant	UCI Acc	UCI F1	WISDM Acc	WISDM F1
Full	94.63	94.66	96.65	95.71
w/o SERA	90.81	86.25	91.69	88.14
w/o ICCA	92.28	90.12	93.86	92.37
SERA 1/4	85.44	81.79	86.32	84.26
SERA 1/2	89.56	88.53	90.17	88.29
SERA full	95.04	95.11	96.86	96.20

Table 6 Theoretical time savings relative to fixed windows (%)

Dataset	1/4	1/2	Full	Avg
UCI-HAR	24.35	49.57	26.08	43.04
WISDM	22.51	51.20	26.29	42.45
KU-HAR	17.28	31.48	51.24	28.69

4.5.2 Workstation Inference and Edge Implications

Table 7 reports end-to-end inference latency and throughput measured on a workstation under a consistent software stack. these results are not from on-device tests but provide a device-agnostic comparison. SERA-HAR achieves a favorable speed–accuracy trade-off, with latency close to a lightweight CNN and clearly lower than recurrent and Transformer baselines.

To approximate real-world deployment, we applied lightweight optimization techniques including INT8 quantization, structured channel pruning, and Conv–BN fusion with fixed-point activations. As shown in Table 8, these techniques reduce

Table 7 Measured latency and throughput on the workstation. Throughput is in samples per second

Model	Latency [ms]	Throughput [sps]
Random Forest	0.72	1389
CNN	1.15	870
LSTM	2.46	412
Transformer	4.39	228
ST-GCN	6.27	136
SERA-HAR	1.79	760

Table 8 Lightweight deployment estimates after quantization and pruning.

Model	Params [M]	Size INT8 [MB]	Est. Lat. [ms]
CNN	1.2	1.2	15.3
LSTM	2.8	2.8	34.7
Transformer	5.4	5.4	55.1
SERA-HAR	3.1	3.1	22.8

parameter size to only a few MB and yield estimated latencies of 20–25 ms on ARM Cortex-A72 CPUs, which is well aligned with typical HAR window strides. The early-exit threshold θ further allows tuning latency to meet a given budget. Although a full hardware deployment remains future work, these device-oriented results provide strong evidence for the feasibility of SERA-HAR on low-power edge platforms.

5 Conclusion

This work presented **SERA-HAR**, a real-time human activity recognition framework that integrates a residual TCN with staged early-exit attention and inter-channel correlation attention (ICCA). Evaluated on UCI-HAR, WISDM, and KU-HAR, it achieved 94.6–96.7% accuracy and up to 95.95% macro-F1. The staged mechanism reduced expected observation time by up to 43%, and measured latency was 1.79 ms with high throughput. Lightweight quantization and pruning further indicated feasibility on ARM Cortex-A72 CPUs, confirming practicality for low-power devices. Beyond benchmarking, these results suggest important practical implications. The early-exit mechanism allows low-latency and energy-efficient recognition, which is particularly valuable for time-sensitive applications such as fall detection, fitness tracking, and mobile health monitoring. Moreover, the lightweight design enables deployment on wearable and smartphone platforms, making continuous and sustainable HAR feasible in real-world daily-life and healthcare scenarios.

In future work, we will extend SERA-HAR to richer sensing modalities, enhance the uncertainty-aware exit policy, and perform large-scale validation on embedded and mobile platforms to strengthen its applicability in real-world deployments.

Author Contributions Zhengjie Li: Conceptualization, Methodology, Software, Experiments, Writing—original draft. Ming Fu: Data curation, Visualization, Writing—review and editing. Baole Fu: Formal analysis, Validation, Writing—review and editing. Yinhua Liu: Supervision, Project administration, Funding acquisition, Writing—review and editing. All authors approved the final manuscript.

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Data Availability UCI-HAR, WISDM, and KU-HAR are publicly available at their respective sources. Processed data and analysis scripts are available upon reasonable request.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing Interests The authors declare no competing interests.

Code Availability Implementation code is available from the corresponding author upon request.

References

1. Attal, F., Mohammed, S., Dedabrishvili, M., Chamroukhi, F., Oukhellou, L., Amirat, Y.: Physical human activity recognition using wearable sensors. *Sensors* **15**(12), 31314–31338 (2015). <https://doi.org/10.3390/s151229858>
2. Lara, O.D., Labrador, M.A.: A survey on human activity recognition using wearable sensors. *IEEE Commun. Surv. Tutor.* **15**(3), 1192–1209 (2013). <https://doi.org/10.1109/SURV.2012.110112.00192>
3. Zhang, S., Li, Y., Zhang, S., Shahabi, F., Xia, S., Deng, Y., Alshurafa, N.: Deep learning in human activity recognition with wearable sensors: A review on advances. *Sensors* **22**(4), 1476 (2022). <https://doi.org/10.3390/s22041476>
4. Picerno, P., Iosa, M., D'Souza, C., Benedetti, M.G., Paolucci, S., Morone, G.: Wearable inertial sensors for human movement analysis: A five-year update. *Expert Rev. Med. Devices* **18**(S1), 79–94 (2021). <https://doi.org/10.1080/17434440.2021.1988849>
5. Huang, Y., Zhou, Y., Zhao, H., Riedel, T., Beigl, M.: A survey on wearable human activity recognition: innovative pipeline development for enhanced research and practice. *Proc. IEEE IJCNN* (2024). <https://doi.org/10.1109/IJCNN60899.2024.10650060>
6. Contoli, C., Freschi, V., Lattanzi, E.: Energy-aware human activity recognition for wearable devices: A comprehensive review. *Pervasive Mob. Comput.* **104**, 101976 (2024). <https://doi.org/10.1016/j.pmcj.2024.101976>
7. Chen, K., Zhang, D., Yao, L., Guo, B., Yu, Z., Liu, Y.: Deep learning for sensor-based human activity recognition: overview,

- challenges, and opportunities. *ACM Computing Surveys* **54**(4), 77 (2021). <https://doi.org/10.1145/3447744>
8. Ferrara, E.: Large language models for wearable sensor-based human activity recognition, health monitoring, and behavioral modeling: A survey of early trends, datasets, and challenges. *Sensors* **24**(15), 5045 (2024). <https://doi.org/10.3390/s24155045>
 9. Sedaghati, N., Ardebili, S., Ghaffari, A.: Application of human activity/action recognition: a review. *Multimedia Tools and Applications* (2025). <https://doi.org/10.1007/s11042-024-20576-2>
 10. Jiang, W., Yin, Z.: Human activity recognition using wearable sensors by deep convolutional neural networks. *Proc. ACM Multimedia (MM'15)*, 1307–1310 (2015). <https://doi.org/10.1145/2733373.2806333>
 11. Ramanujam, E., Perumal, T., Padmavathi, S.: Human activity recognition with smartphone and wearable sensors using deep learning techniques: A review. *IEEE Sensors J.* **21**(12), 13029–13040 (2021). <https://doi.org/10.1109/JSEN.2021.3069927>
 12. Ordóñez, F.J., Roggen, D.: Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition. *Sensors* **16**(1), 115 (2016). <https://doi.org/10.3390/s16010115>
 13. Park, H., Kim, N., Lee, G.H., Choi, J.K.: MultiCNN-FilterLSTM: Resource-efficient sensor-based human activity recognition in IoT applications. *Futur. Gener. Comput. Syst.* **139**, 196–209 (2023). <https://doi.org/10.1016/j.future.2022.09.024>
 14. Hu, K., Jin, J., Zheng, F., Weng, L., Ding, Y.: Overview of behavior recognition based on deep learning. *Artif. Intell. Rev.* **56**, 1833–1865 (2023)
 15. Zhang, H., Xiao, Z., Wang, J., Li, F., Szczerbicki, E.: A novel IoT-perceptive human activity recognition (HAR) approach using multihead convolutional attention. *IEEE Internet Things J.* **7**(2), 1072–1080 (2020)
 16. Lattanzi, E., Contoli, C., Freschi, V.: Do we need early exit networks in human activity recognition? *Eng. Appl. Artif. Intell.* **121**, 106035 (2023)
 17. Sharen, H., Anbarasi, L.J., Rukmani, P., Gandomi, A.H., Neeraja, R., Narendra, M.: WISNet: A deep neural network based human activity recognition system. *Expert Syst. Appl.* **258**, 124999 (2024). <https://doi.org/10.1016/j.eswa.2024.124999>
 18. Zhang, H., Xu, L.: Multi-STMT: Multi-level network for human activity recognition based on wearable sensors. *IEEE Trans. Instrum. Meas.* **73**, 2508612 (2024). <https://doi.org/10.1109/TIM.2024.3365155>
 19. Dahou, A., Al-qaness, M.A.A., Abd Elaziz, M., Helmi, A.: Human activity recognition in IoT applications using arithmetic optimization algorithm and deep learning. *Measurement* **199**, 111445 (2022). <https://doi.org/10.1016/j.measurement.2022.111445>
 20. Dhekane, S.G., Ploetz, T.: Transfer learning in sensor-based human activity recognition: A survey. *ACM Computing Surveys* **57**(8), 205 (2025). <https://doi.org/10.1145/3717608>
 21. Ganesha, H.S., Gupta, R., Gupta, S.H., Rajan, S.: Few-shot transfer learning for wearable IMU-based human activity recognition. *Neural Comput. Appl.* **36**, 10811–10823 (2024). <https://doi.org/10.1007/s00521-024-09645-7>
 22. Chen, X., Zhou, X., Sun, M., Wang, H.: Temporal contrastive learning for sensor-based human activity recognition: A self-supervised approach. *IEEE Sens. J.* **25**(1), 1839–1847 (2025). <https://doi.org/10.1109/JSEN.2024.3491933>
 23. Xiao, Z., Tong, H., Qu, R., Xing, H., Luo, S., Zhu, Z., Song, F., Feng, L.: CapMatch: Semi-supervised contrastive transformer capsule with feature-based knowledge distillation for human activity recognition. *IEEE Transactions on Neural Networks and Learning Systems* **36**(2), 2690–2704 (2025). <https://doi.org/10.1109/TNNLS.2025.3556317>
 24. Kia, Z., Yadollahzaeh-Tabari, M., Motameni, H.: Human activity recognition by body-worn sensor data using bi-directional generative adversarial networks and frequency analysis techniques. *J. Supercomput.* **81**, 342 (2025). <https://doi.org/10.1007/s11227-024-06743-0>
 25. Xu, H., Li, Y., Dong, Q., Liu, L., Li, J., Zhang, J., Wang, B.: Random code radar with range–time–frequency points and improved pointconv network for through-wall human action recognition. *IEEE Sens. J.* **25**(8), 13719–13730 (2025). <https://doi.org/10.1109/JSEN.2025.3548121>
 26. Oishi, N., Birch, P., Roggen, D., Lago, P.: WIMUSim: Simulating realistic variabilities in wearable IMUs for human activity recognition. *Frontiers in Computer Science* **7**, 1514933 (2025). <https://doi.org/10.3389/fcomp.2025.1514933>
 27. Ashry, S., Ogawa, T., Gomaa, W.: CHARM-Deep: Continuous human activity recognition model based on deep neural network using IMU sensors of smartwatch. *IEEE Sens. J.* **20**(15), 8757–8766 (2020). <https://doi.org/10.1109/JSEN.2020.2985374>
 28. Mekruksavanich, S., Jitpattanakul, A.: Hybrid convolution neural network with channel attention mechanism for sensor-based human activity recognition. *Sci. Rep.* **13**, 12067 (2023). <https://doi.org/10.1038/s41598-023-39080-y>
 29. Waghumbare, A., Singh, U.: DIAT-separable-CNN-ECA-HARNet: A lightweight and explainable model for efficient human activity recognition. *SIVIP* **19**, 245 (2025). <https://doi.org/10.1007/s11760-025-03838-w>
 30. Wen, F., Zhang, D., Wang, J., Sun, Y.: An adaptive attention-based hybrid deep model for sensor-based human activity recognition. *J. Ambient. Intell. Humaniz. Comput.* **14**(5), 1234–1247 (2023)
 31. Liandana, M., Hostiadi, D.P., Hendrawan, N.R., Pradipta, G.A., Wulaning Ayu, P.D.: Enhanced human activity recognition (HAR):leveraging sub-window techniques and feature ratios from triaxial accelerometer data. *International Journal of Intelligent Engineering and Systems* **17**(5), 481–491 (2024). <https://doi.org/10.22266/ijies2025.0229.34>
 32. Dehghani, A., Sarbishei, O., Glatard, T., Shihab, E.: A quantitative comparison of overlapping and non-overlapping sliding windows for human activity recognition using inertial sensors. *Sensors* **19**(22), 5026 (2019). <https://doi.org/10.3390/s19225026>
 33. Anguita, D., Ghio, A., Oneto, L., Parra, X., Reyes-Ortiz, J.L.: A public domain dataset for human activity recognition using smartphones. *Proc. ESANN*, 437–442 (2013)
 34. Kwapisz, J.R., Weiss, G.M., Moore, S.A.: Activity recognition using cell phone accelerometers. *SIGKDD Explorations* **12**(2), 74–82 (2011). <https://doi.org/10.1145/1964897.1964918>
 35. Kim, H., Lee, S., Kwon, S.: A dataset for user activity recognition using body-mounted sensors with multiple modalities. *Sensors* **18**(2), 565 (2018). <https://doi.org/10.3390/s18020565>
 36. Ullah, H., Munir, A.: Human activity recognition using cascaded dual attention CNN and bi-directional GRU framework. *Journal of Imaging* **9**(7), 130 (2023). <https://doi.org/10.3390/jimaging9070130>
 37. Lalwani, P., amasamy, G.R.: Human activity recognition using a multi-branched CNN-BiLSTM-BiGRU model. *Applied Soft Computing* **143**, 111344 (2024). <https://doi.org/10.1016/j.asoc.2024.111344>

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